



Write down the inequality in y shown by the number line.

$$-2.5 \leq y < 4 \quad [2]$$

- 8 Hadi buys a dishwasher.
He can either pay a single payment of \$980 or he can pay using a monthly plan.
The monthly plan is 20% of \$980 **plus** 12 payments of \$75.25.

(a) Hadi uses the monthly plan.

Calculate the total amount Hadi pays.

$$= 980 \times 0.2$$

$$= \$196$$

$$= 12 \times 75.25$$

$$= \$867$$

$$= 196 + 867 = \$1063$$

$$\$ \dots\dots\dots 1063 \dots\dots\dots [2]$$

(b) Find the percentage increase in the cost using the monthly plan compared to a single payment.

$$= \frac{1063 - 980}{980} \times 100$$

$$= 8.47 \%$$

$$\dots\dots\dots 8.47 \dots\dots\dots \% [2]$$

- 9 In a sale, the original price of a sewing machine is reduced by \$38.
This is an 8% reduction in the original price.

Work out the original price of the sewing machine.

$$= \begin{array}{r} 38 \quad \% \quad 8 \\ \times \quad \% \quad 100 \\ \hline \end{array}$$

$$= \frac{3800}{8} = \frac{8}{8} \times$$

$$475 = x$$

$$\$ \dots\dots\dots 475 \dots\dots\dots [2]$$